

EXECUTIVE MEMO

P&C Pricing Diagnostic

This memo summarizes a synthetic P&C pricing diagnostic designed to help insurance teams identify pricing, underwriting, risk classification, and data quality review priorities before committing to a full pricing review.

PURPOSE

Translate model outputs and portfolio diagnostics into a short list of review priorities that can be discussed by pricing, underwriting, and analytics teams.

1. Executive summary

The diagnostic reviews claim frequency, loss ratio pressure, benchmark model calibration, and observed-versus-expected claim behavior across portfolio segments.

The objective is not to produce final rate indications. The objective is to identify where deeper pricing or underwriting work is most likely to create value.

Primary signal

Segment-level frequency and loss ratio indices identify areas that may be performing worse than the portfolio average.

Model support

A benchmark Poisson frequency GLM supports expected claim estimates and observed-versus-expected diagnostics.

Decision use

Findings are organized into review priorities, not automatic pricing actions.

2. Scope reviewed

- Portfolio experience by segment
- Observed claim frequency
- Loss ratio and loss ratio index
- Benchmark frequency model outputs
- Observed-versus-expected claims
- Calibration by expected-risk decile
- Model specification comparison
- Executive visual outputs

How this diagnostic is used in practice

Start with one portfolio slice

Review one book, program, territory cluster, or segment group before expanding the workflow.

Prioritize review areas

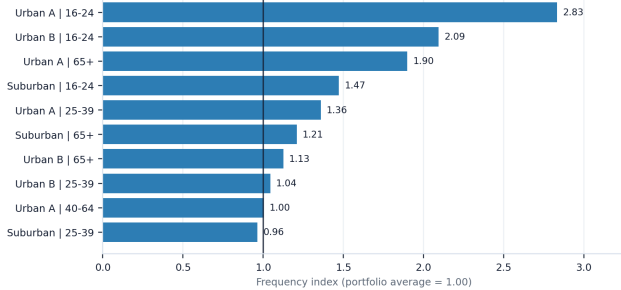
Identify where pricing, underwriting, risk classification, or data quality review is most likely to matter.

Decide the next level of work

Determine whether deeper modeling, monitoring, or pricing action is justified.

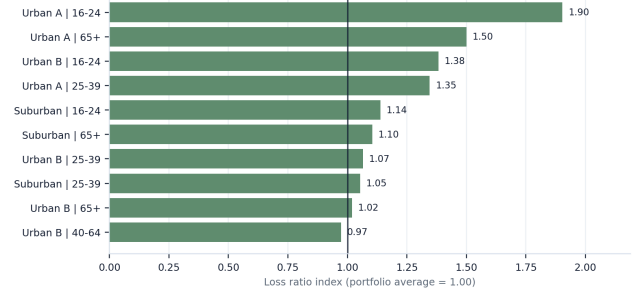
3. Elevated frequency segments

The frequency index compares each segment's observed claim frequency against the portfolio average. Values above 1.00 indicate higher frequency than the portfolio average.



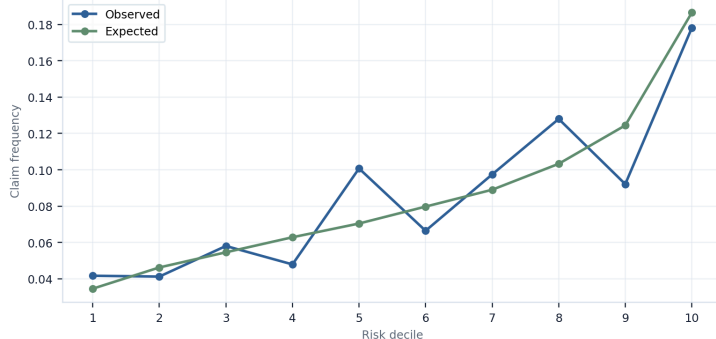
4. Loss ratio pressure

The loss ratio index compares each segment's loss ratio against the total portfolio loss ratio. Values above 1.00 indicate worse loss ratio performance than the portfolio average.



5. Model validation note

Policies are sorted by model-expected frequency and grouped into ten equally sized risk deciles. Higher deciles should generally show higher observed frequency.



6. Recommended next steps

- Validate exposure, premium, claims, and underwriting fields.
- Review high-frequency and high-loss-ratio segments.
- Separate frequency and severity drivers.
- Identify 3–5 pricing, underwriting, or risk classification priorities.
- Decide whether deeper modeling or monitoring is justified.

7. Limitations

- Synthetic sample only.
- Not a final pricing indication.
- No expense, reinsurance, capital, or regulatory treatment.
- Requires real data validation before production use.

Interpretation note

These findings are prioritization signals. They should not be interpreted as automatic rate changes without reviewing credibility, underwriting context, severity contribution, data quality, and business constraints. This commercial sample uses synthetic data and demonstrates workflow structure, not final actuarial pricing advice.

What the diagnostic produces

Pricing review priorities

Segments with elevated frequency, loss ratio pressure, or observed-versus-expected deterioration.

Underwriting review priorities

Risk classifications, eligibility rules, coverage terms, or segment definitions that may deserve closer review.

Monitoring priorities

Segments, fields, or model diagnostics that should be tracked before deeper pricing work is approved.